

Lab Assignment No.	1
Title	Problem Statement – Real estate agents want help to predict the house price for regions in the USA. He gave you the dataset to work on and you decided to use the Linear Regression Model. Create a model that will help him to estimate what the house would sell for. URL for a dataset: https://github.com/huzaifsayed/Linear-Regression-Model-for-House-Price-Prediction/blob/master/USA_Housing.csv
Roll No.	
Class	BE (AI&DS)
Date of Completion	
Subject	Computer Laboratory-IV: Deep Learning
Assessor's Sign	

ASSIGNMENT No: 01

Title:-

Implement a Linear Regression Model to predict house prices for regions in the USA using the provided dataset.

Objective:

Develop a model to estimate house prices based on relevant features using Linear Regression.

Outcome:

- Apply Linear Regression in a real-world scenario.
- Understand the implementation of a regression model for house price prediction.

Software Requirements:

- Python (3.x recommended)
- Jupyter Notebook or any Python IDE

Hardware Requirements:

A machine with sufficient RAM and processing power for model training (8GB RAM recommended)

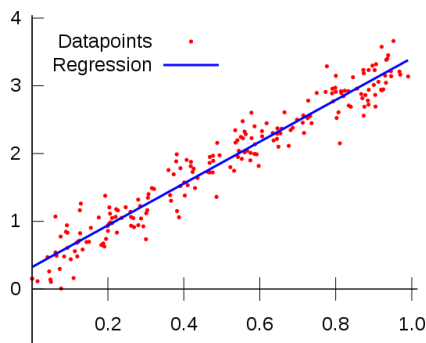
Prerequisites:

- Basic understanding of Python programming
- Familiarity with the concepts of supervised learning

Dataset:

https://github.com/huzaihsayed/Linear-Regression-Model-for-House-Price-Prediction/blob/master/USA_Housing.csv

Linear Regression Model:



Libraries or Modules used:

- NumPy
- pandas
- scikit-learn
- Matplotlib
- Seaborn

Theory:

Linear Regression:

What is Linear Regression?

Linear regression is a **machine learning algorithm** used for predicting numerical values. It finds a **straight-line relationship** between a dependent variable (what you want to predict) and one or more independent variables (the factors that influence it).

Example

Imagine you want to predict **a student's exam score** based on the **number of hours they study**. The more they study, the higher the score—this is a linear relationship.

The equation for linear regression is:

$$Y = WX + b$$

Where:

- **Y**= Predicted value (e.g., exam score)
- **X**= Input feature (e.g., study hours)
- **W**= Weight (determines how much X affects Y)
- **b**= Bias (adjusts the prediction up or down)

Types of Linear Regression

1. What is Linear Regression?

Linear regression is a statistical method used to model the relationship between a dependent variable and one independent variable. It aims to establish a linear relationship between these variables and can be used for both prediction and understanding the nature of the relationship.

Mathematical Equation

The mathematical representation of simple linear regression is: $Y = C_0 + C_1X + e$

where,

- Y : Dependent Variable (target variable)
- X : Independent Variable (input variable)
- C_0 : Intercept (value of Y when $X=0$)
- C_1 : Slope of line
- e : Error term

Step-by-Step Algorithm

Step 1: Collect Data

Gather values for X (input) and Y (output).

Step 2: Calculate Mean of X and Y

$$\bar{X} = \frac{1}{N} \sum X_i, \quad \bar{Y} = \frac{1}{N} \sum Y_i$$

where N is the number of data points.

Step 3: Compute the Slope (C_1)

$$C_1 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

Step 4: Compute the Intercept (C_0) :

$$C_0 = \bar{Y} - C_1\bar{X}$$

Step 5: Construct the Regression Equation:

$$Y = C_0 + C_1X$$

Step 6: Make Predictions

For a new input X_{new} predict Y_{new} using:

$$Y_{new} = C_0 + C_1X_{new}$$

Step 7: Evaluate the Model

Calculate **Mean Squared Error (MSE)** to measure accuracy:

$$MSE = \frac{1}{N} \sum (Y_i - \hat{Y}_i)^2$$

2. What is Multiple Regression?

Multiple regression is an extension of simple linear regression. It models the relationship between one dependent variable and two or more independent variables. The primary purpose is to understand how the dependent variable changes as the independent variables change.

Mathematical Equation

The mathematical representation of multiple regression is:

$$Y = C_0 + C_1X_1 + C_2X_2 + C_3X_3 + \dots + C_nX_n + e$$

where,

- **Y:** Dependent Variable (target variable)
- **$X_1, X_2, X_3, \dots, X_n$:** Independent Variable (input variable)
- **C_0 :** Intercept (value of Y when $X=0$)
- **$C_1, C_2, C_3, C_4, C_5, \dots, C_n$:** Slope of line
- **e:** Error term

Step-by-Step Algorithm

Step 1: Collect Data

Gather the data consisting of multiple independent variables ($X_1, X_2, \dots, X_n$) and the dependent variable Y .

Step 2: Normalize or Standardize Data

(Optional but recommended)

- Scale the features $X_1, X_2, \dots, X_n$ to improve performance and stability of the algorithm.
- Common methods: **Standardization** (mean = 0, variance = 1) or **Min-Max scaling**.

Step 3: Calculate the Mean of Each Variable

Calculate the mean for each independent variable $\bar{X}_1, \bar{X}_2, \dots, \bar{X}_n$ and the dependent variable $\bar{Y}$

Step 4: Compute the Coefficients ($C_1, C_2, \dots, C_n$)

Using the **Normal Equation** or **Gradient Descent**:

The **Normal Equation** approach is used when we have a closed-form solution:

$$C = (X^T X)^{-1} X^T Y$$

where:

X is the matrix of independent variables (with an added column of 1s for C_0 for the intercept term)

Y is the dependent variable vector

C is the vector of coefficients $[C_0, C_1, \dots, C_n]$

Alternatively, **Gradient Descent** can be used iteratively to find the best coefficients.

Step 5: Calculate the Intercept C_0

Once the coefficients $C_1, C_2, \dots, C_n$ are computed, calculate the intercept C_0 :

$$C_0 = \bar{Y} - \sum_{i=1}^n C_i \bar{X}_i$$

Step 6: Build the Regression Equation

Now, construct the multiple linear regression models: $Y = C_0 + C_1X_1 + C_2X_2 + \dots + C_nX_n$

Step 7: Make Predictions

For new input values $X_1, X_2, \dots, X_n$ predict the corresponding output Y_{new} using the equation:

$$Y_{new} = C_0 + C_1X_1 + C_2X_2 + \dots + C_nX_n$$

Step 8: Evaluate the Model

Evaluate the model's performance using metrics such as:

Mean Squared Error (MSE):

$$MSE = \frac{1}{N} \sum_{i=1}^N (Y_i - \hat{Y}_i)^2$$

R-Squared (R^2): Measures how well the independent variables explain the variance in the dependent variable.

Linear Regression vs Multiple Regression:

Parameter	Linear (Simple) Regression	Multiple Regression
Definition	Models the relationship between one dependent and one independent variable.	Models the relationship between one dependent and two or more independent variables.
Equation	$Y = C_0 + C_1X + e$	$Y = C_0 + C_1X_1 + C_2X_2 + C_3X_3 + \dots + C_nX_n + e$
Complexity	It is simpler to deal with one relationship.	More complex due to multiple relationships.
Use Cases	Suitable when there is one clear predictor.	Suitable when multiple factors affect the outcome.
Assumptions	Linearity, Independence, Homoscedasticity, Normality	Same as linear regression, with the added concern of multicollinearity.
Visualization	Typically visualized with a 2D scatter plot and a line of best fit.	Requires 3D or multi-dimensional space, often represented using partial regression plots.
Risk of Overfitting	Lower, as it deals with only one predictor.	Higher, especially if too many predictors are used without adequate data.
Multicollinearity Concern	Not applicable, as there's only one predictor.	A primary concern; having correlated predictors can affect the model's accuracy and interpretation.
Applications	Basic research, simple predictions, understanding a singular relationship.	Complex research, multifactorial predictions, studying interrelated systems.

Applications:

1. House price prediction
2. Sales forecasting
3. Weather prediction
4. Stock market trends

Conclusion: Thus Implemented a Linear Regression Model to predict house prices for regions in the USA using the provided dataset.